## Activity 4.1 – Hazard Detection and Forwarding Unit

### 1.Design Approach:

Designed a hazard detection and forwarding unit for a pipelined RISC-V processor using Verilog HDL. Implemented load-use hazard detection and forwarding selection logic to handle data dependencies between pipeline stages.

### 2.Verification Steps:

- Created a separate testbench.
- Applied register dependency conditions.
- Tested load-use hazard detection.
- Verified stall signal generation.
- Tested forwarding paths.
- Checked forwarding selection signals.
- Simulated using ModelSim.

### 3.Results:

The testbench checks data hazard detection, stall generation and forwarding selection. PASS/FAIL messages are displayed.

### 4.Issues Faced:

Register dependency conditions must be checked correctly. Forwarding priority must be maintained. Load-use hazards require proper stall generation.

### 5.Conclusion:

Successfully designed and verified a hazard detection and forwarding unit using Verilog HDL.